



Operations Research

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Exercise Sheet 10

Demo Exercises

Exercise D10.1. *Topological Properties and Convex Sets*

Consider the following two sets in $\mathbb{R}^2$:

$$X_1 = \{x \in \mathbb{R}^2 \mid \|x\|_2 < 1\}$$

$$X_2 = \{x \in \mathbb{R}^2 \mid x_1 + x_2 \leq 1, x_1 \geq 0, x_2 \geq 0\}$$

- (a) Determine for both X_1 and X_2 whether they are open, closed, bounded, and/or compact. Justify your answers mathematically.
- (b) Prove rigorously that X_2 is a convex set.

Exercise D10.2. *Unconstrained Optimization*

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be given by $f(x, y) = x^2y + y^3 - 3y$.

- (a) Compute the gradient $\nabla f(x, y)$ and the Hessian matrix $H_f(x, y)$.
- (b) Determine all stationary points of f .
- (c) Classify each stationary point (local minimum, local maximum, or saddle point) by explicitly calculating the eigenvalues of the Hessian matrix at each point.

Exercise D10.3. *KKT-conditions*

- (a) (4P) Let $f : \mathbb{R}^n \rightarrow \mathbb{R}, g : \mathbb{R}^n \rightarrow \mathbb{R}^m, h : \mathbb{R}^n \rightarrow \mathbb{R}^p$.
Consider the following constrained non-linear optimization problem.

$$\min f(x) \text{ s.t. } g(x) \leq 0, h(x) = 0$$

State the KKT-conditions.

- (b) (1P) How do the KKT conditions change for a maximization problem?
You only need to write down the changes compared to (a).

For the following exercises let

$$f(x, y) = \exp(2x^2 + 4y^2 + 24y) \\ h(x, y) = x^2 + (y + 3)^2 - 1.$$

And let the problem (P) be given by

$$(P) \quad \min f(x, y), \text{ s.t. } h(x, y) = 0$$

- (c) (3 P) Show that f is convex. You may use without proof that the composition $f_1 \circ f_2$ is convex if f_1 and f_2 are convex and f_1 is strictly increasing.
- (d) (1P) Show that Slater's condition is violated for (P).
- (e) (3P) Compute the gradients of f and h .
- (f) (2P) Show that $\nabla h \neq 0$ for all feasible points. This implies that all feasible points are regular.

You may assume without proof that the feasible region is compact, hence the continuous function f attains both a global minimum and maximum. Furthermore, assume that there exists four KKT-points in total (two for maximization and two for minimization), which are given by z_1, z_2, z_3 and z_4 .

- (g) (3P) Describe how you would determine the global minima and maxima of f under the constraint $h = 0$. Justify your approach.
You do not need to explicitly compute the maximizers and minimizers.